

Nathaniel Lee

nwlee@utexas.edu ♦ (330) 641-6116

 LinkedIn  Portfolio Website

WORK EXPERIENCE

General Motors

June 2022 – Sep 2024

Software Developer / Systems Engineer

Warren, MI

- Administered enterprise NAS and SAN storage infrastructure across 100+ global sites in 30+ countries, ensuring high availability and compliance for mission-critical operations.
- Managed the lifecycle of IT service requests via our internal ticketing system, utilizing ITIL frameworks for Incident, Problem, Change, and Configuration Management.
- Provided critical on-call support during non-production hours to execute OS upgrades, hardware installations, and data migrations, coordinating closely with enterprise vendors including Dell/EMC and IBM.
- Managed user access control, modified group permissions, and provisioned storage capacity across both Windows (SMB) and UNIX/Linux (NFS) environments.
- Authored and executed UNIX shell commands via SSH to complete daily administrative tasks, including data replication, decommissioning legacy files, and managing storage quotas.
- Engineered Python batch scripts to parse over 10,000 service tickets, automatically generating dashboards that visualized team performance metrics for executive review.
- Mitigated operational risk by strictly adhering to change management practices, drafting implementation, verification, and backout command sequences for formal change requests within the ITSM platform.
- Identified operational bottlenecks and streamlined workflows by integrating ticketing systems with Python/Django automation tools via REST APIs, successfully eliminating hundreds of manual labor hours annually.

Ohio State University Risk and Emergency Management

Oct 2021 – April 2022

Access Control Assistant

Columbus, OH

- Managed campus-wide access control systems by diagnosing hardware failures and writing custom batch scripts to automate maintenance logging and reporting for university facilities.

Ohio State University Center for Electron Microscopy and Analysis

April 2021 – Aug 2021

Research Assistant

Columbus, OH

- Accelerated materials characterization research by implementing 3D reconstruction algorithms in CryoSPARC, reducing sample processing time from days to hours through optimized image analysis.

EDUCATION

University of Texas at Austin

Dec, 2027

MS, Computer Science

Austin, TX

- 3.7 GPA

Ohio State University

May, 2022

BS, Computer Science and Engineering

Columbus, OH

- Dean's List; 3.41 GPA

PROJECTS

Minecraft Forge Open-Source Collaboration and Development

- Collaborated on open-source Java development using Spring Boot and the Forge API, implementing core gameplay features and resolving critical bugs to enhance system stability and performance for the modding community.

Quality of Wine Prediction Models

- Architected predictive neural networks (CNNs, RNNs) in Python using TensorFlow and Keras to analyze complex meteorological datasets, successfully correlating regional weather patterns with wine quality metrics.

Spotify Playlist Recommender Web Application

- Engineered a full-stack chatbot application integrating Amazon Lex, Django, and React, utilizing OAuth 2.0 for secure authentication to generate personalized Spotify API playlists based on user sentiment analysis.

SKILLS

Java, Python, C, C++, C#, Ruby, MATLAB, R, SQL, JavaScript, HTML, CSS, PyTorch, TensorFlow, Spring Boot, Django, Kotlin, React, Angular, Bootstrap, jQuery, PostgreSQL, SQLite, MongoDB, Node.js, Git, Docker, Azure DevOps CI/CD, Linux/UNIX, Batch/Shell Scripting